

The regional employment implications of a net-zero economy in Costa Rica under uncertainty

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Abstract

This study estimates job creation from Costa Rica's transition to net-zero emissions by 2050, contrasting it with a business-as-usual scenario (BAU) using employment multipliers from Costa Rica's input–output matrix. The model is soft-linked with a bottom-up technology-rich regional model of the CLEW sectors in Costa Rica. A robust decision-making-inspired approach is taken to assess the effects of uncertainty on the estimations. The study addresses a literature gap in modeling subnational employment while accounting for long-term uncertainty and allows to identify economic activities affected by the transformation, so that policies regarding a just transition can be properly developed and stakeholders incentivized to take active participation in the process. Under baseline assumptions, a net-zero economy would have 135 thousand more jobs by 2050 than the BAU, equivalent to 7% of the employed population in 2017. Energy-related jobs would triple by mid-21st century, and regions capitalizing on renewables would perceive the highest gain. When considering uncertainty, the net jobs range between 35–750 thousand. Since net positive outcomes in terms of jobs are feasible, global governance should focus efforts on policies that enhance these.

Keywords: Decarbonization; Environmental policy; Input–output; Employment multipliers; Costa Rica

1. Introduction

Costa Rica aims to become a net-zero economy by 2050 by implementing its National Decarbonization Plan 2018–2050 (NDP; GCR, 2018), Nationally Determined Contributions (NDC; GCR, 2020), and Territorial Economic Strategy for an Inclusive and Decarbonized Economy 2020–2050 (TES; GCR, 2021). These plans pursue electrifying the transport and industry sectors with a 100% renewable electricity system by

2050. They seek increased coverage in waste management and enforce sustainable practices in construction and agriculture. It is estimated that decarbonizing the Costa Rican economy would generate 41 billion USD in economic benefits by 2050 (Groves et al., 2020), equivalent to 64% of 2019 GDP; about half of these arising in the energy sector (Godínez-Zamora et al., 2020). However, besides relative cost differences between a decarbonization and a business-as-usual scenario, job creation estimates are relevant for Costa Rica, since it faces high unemployment levels, particularly when compared to other OECD countries (OECD, 2023).

For Latin America and the Caribbean, structural changes in production and consumption patterns, aligned with net-zero emissions by mid-century, are estimated to result in 15 million

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net jobs by 2030 (Saget et al., 2020) (*i.e.*, additional jobs with decarbonization than without). A transition linked to clean energy and electric mobility in the region would create more than 35 million net jobs by 2050 (PNUMA, 2020). These numbers are compatible with employment factor estimations for renewable investment, which produces 5 more jobs per 1 million USD than fossil fuel investment (Garrett-Peltier, 2017).

The impacts of low-emission policies on growth and employment have been challenging to quantify (Bowen and Kuralbayeva, 2015). With regards to production, economic growth has been a key factor leading to the growth of global carbon emissions (Li et al., 2021). The intensity of the relation between growth and carbon emissions, though, depends on different variables, such as income inequality for high income countries (Wang et al., 2023b). Additionally, it has been estimated that the lower the income level in the countries, the more unstable their decoupling state can be (Wang et al., 2023a). Therefore, depending on the characteristics of each country the impacts of climate policies can have different effects on growth. Costa Rica in particular has been found to be in a state of decoupling where carbon emissions per capita increase with GDP per capita and decrease with human capital and resource rent (Wang et al., 2023d).

Regarding employment, recently job creation estimates have focused on the energy transition at a global scale (IRENA and ILO, 2021; Ram et al., 2022), with some state or nation-wide examples for California (DeShazo et al., 2018), Italy (Dell'Anna, 2021), and Canada (Bridge, 2017; Mertins-Kirkwood, 2018). For example, direct energy jobs in the power, heat, transport, and desalination sectors can increase from 57 million world-wide in 2020 to 134 million in 2050 (Ram et al., 2022). For Latin America, some studies have incorporated firm-level surveys to estimate jobs per unit of investment in Bolivia, Chile, and Uruguay (Ravillard et al., 2021). Economy-wide estimates are only available for Argentina (ILO, 2019), but are not strictly aligned with net-zero economy transitions.

The aforementioned studies are missing two fundamental insights for policymakers. First, since decarbonization will unfold unevenly within countries (DDPP, 2015; Sonja and Harald, 2018), with some provinces benefiting more than others, incorporating subnational estimations can harness support for decarbonization and managing impacts in affected regions. Second, long-term estimates are based on best guesses about the future (Ackerman et al., 2009; Beck and Krueger, 2016; Naimoli and Ladislav, 2020; Pindyck, 2017), but incorporating robust decision-making methods can shed light on the causes of higher or lower job creation levels. These two additional insights can support the sustained commitment (Li and Strachan, 2019; Waisman, 2018) necessary to deploy the net-zero emission plans. Moreover, identifying the specific economic activities that will gain or lose net jobs by 2050 allows to plan, develop and take actions related to the formation of human capital and capabilities in a timely manner, so that the positive effects of the transition can be maximized and the negative effects minimized.

This study presents an estimation of the jobs that will result from the transition to a net-zero economy by 2050 in Costa Rica. Costa Rica has always pioneered in climate action, and was one of the first countries to develop a proper decarbonization plan; although not one of the highest global-polluters, its leadership in this topic is important so other countries, particularly in Latin American, follow. This estimation complements the existing knowledge on the economic benefits of this decarbonization. A subnational input–output model is used to create two scenarios: business-as-usual and net-zero emissions. This model is soft-linked with a bottom-up technology-rich regional model of the climate, land, energy, and water (CLEW) sectors in Costa Rica. The soft-link ensures the compatibility of the input–output results with economic activity demands complying with net-zero by 2050. Finally, the effects of uncertainties are studied by implementing a robust decision-making analysis.

2. Methods and data

2.1. Leontief multipliers

Leontief multipliers (Leontief, 1986) are used to quantify the impact of the transition to a net-zero economy on employment. The input–output approach and Leontief multipliers have been used historically in environmental analysis (Guilhoto, 2021). In academic literature, the use of this methodology has focused on carbon analysis for cities (Ramaswami et al., 2021), economic impacts of major events (Collins et al., 2012), impacts of public spending (Morgan et al., 2017), effects of disasters (Okuyama and Santos, 2014) and changes in technologies such as bio-fuels (Allan, 2015). Additionally, the multipliers have been used (Durán Lima and Banacloche, 2022) and recommended (GAIN, 2017) for production and employment estimations specific to Latin America under new policies.

Augmented Leontief multipliers, which indicate the impact changes on the demand have on the economic activities' output and employment (Agriculture Organization of the United Nations and International Labour Organization, 2020; GAIN, 2017) are calculated using the regional Costa Rican input–output matrix. These multipliers allow to trace all the activities that are affected by one change (or several) in demand, and to estimate their magnitude, which makes them appropriate to the specific research question. Calculations on variations of job levels related to specific economic activities allows to map the changes in the dynamics of the economy. This mapping proves important when developing policies that affect directly the labor market in light of the transformation of the economy. The methodology explained in Miller and Blair (2009) is initially followed to obtain the Leontief inverse matrix and therefore, Leontief multipliers.

The multipliers obtained from the Leontief inverse matrix, also known as type I multipliers, only estimate the direct and indirect effects of changes on the demand. Type I multipliers solely focus on the structure of the production system, the relationships between sectors, the impacts of final demand and take the behavior of the final demand as exogenous. Using these, direct and indirect jobs can be estimated.

Direct jobs are positions created directly within the industry under consideration, such as manufacturing in the renewable energy sector. Indirect jobs, however, are those generated in related industries due to the demand created by the primary industry, like jobs in supply sectors supporting renewable energy production. For instance, indirect jobs in the renewable energy sector might include the maintenance services industry, where activities such as maintenance for the equipment and machinery are required in power plants, and they exist due to the demands of the renewable energy industry (see [Tourkolias and Mirasgedis, 2011](#)).

Augmented Leontief multipliers, or type II multipliers, were also calculated. These multipliers include information on consumption and expenditure, so that the induced effects generated by increases in income, associated with induced employment, are also considered. Given those characteristics, induced jobs can be estimated. The induced jobs result from the personal spending of the workers that gained the direct and indirect jobs; in other words, are the result of the dynamization of the economy.

The augmented multipliers were obtained by including households taking the domestic demand only, that includes information on consumption and total labor income. Furthermore, we followed [Ernst and Sarabia \(2015\)](#) to calculate the direct, indirect, and induced effects on employment by using type I and type II multipliers and the employment coefficient by sector, *i.e.*, the employment in sector j divided into the total output of sector j .

Commonly, induced effects (type II) multipliers overestimate the impacts of changes in demands due to the rigid assumptions about household income and expenditure behavior, so that a realistic estimate of these is to meet the direct and indirect (type I) multipliers halfway ([Cassar, 2015](#)). The overestimation of the induced effects is corrected by using the adjusted induced effects, which are the average between the induced effects and the sum of the direct and indirect effects.

2.2. Subnational estimation

Costa Rica's cantonal input–output matrix, produced by the Central Bank of Costa Rica ([Brenes-Soto et al., 2021](#)) is used. This matrix maps the economic linkages of 144 economic activities across the 81 cantons in which the country is administratively divided. Taking the cantonal matrix, we generate a regional matrix, groups the cantons into the six planning regions of the country.

This regional matrix approach is taken in order to apply shocks more precisely and to evaluate the impact of the transformation at a more disaggregated level. The use of this matrix entailed both strengths and limitations. On one hand, the availability of the disaggregated data at a cantonal level is optimal if the shocks are heterogeneously distributed throughout the regions. However, using this matrix implied the handling of an input–output matrix full of zeros, as not all the cantons of the country carry out or interact with all the activities. This implied that both the technical coefficients and Leontief

inverse's coefficients yield very high figures or present errors when inverting the matrix. To solve this problem further, the different economic activities were grouped by sectors. Particularly, the 144 economic activities included in the Costa Rican input–output matrix were grouped into 15 sectors, following the International Standard Industrial Classification of All Economic Activities (ISIC).

2.3. Scenario estimation

We estimated the effect on jobs by contrasting scenarios, taking both the national and regional input–output matrix of Costa Rica's, collected by the Central Bank of Costa Rica in 2017. Two are created: a business-as-usual (BAU) scenario and a net-zero scenario. The main difference between the scenarios is the demand for each economic activity through time. These demands are relevant, since are the ones multiplied by the coefficients to estimate employment.

From the contrasting of scenarios, a key metric to assess the results is derived: *net jobs*, *i.e.*, the difference in the number of estimated jobs in a net-zero scenario and a BAU. A positive difference means the net-zero scenario would produce more jobs than the BAU and a negative difference means the opposite. Zero net jobs mean that the net-zero transition and the BAU scenarios result in equal number of jobs. Details on the assumptions regarding the demands for each scenario are presented in the following subsections.

2.3.1. BAU scenario

The BAU scenario assumes no policies or actions are taken to reach a net-zero economy, *i.e.*, assumes the economy follows its historical trends. It is assumed that the relations between sectors remain constant. Taking the GDP at 2017 prices by economic activity time series (1991–2021) from the Central Bank of Costa Rica, the linear growth trends of each of the productive sectors are calculated. With these trends, the changes in the final demands are estimated and the increase in total production and jobs by 2050 for the scenario is projected. The average year-on-year growth rate of the economic activities is around 1.8%, although the range goes between 0.5% and 4.7%.

The selected time frame (1991–2021) considers several economic events that have affected the trend of different sectors. For example, in 1997, Intel, processor and microprocessor producer, opened up fabric in Costa Rica. This led to other multinationals setting up operations in the country, which translated to high levels of foreign direct investments. In the 2000s, specialization of agricultural products took place, such as luxury coffee and pineapple, affecting exports and growth. Governmental policies, such as the creation of a national 'brand', had positive effects on tourism, which continues to grow. Trends of significant growth in several sectors would reflect as improvements in prosperity in the estimations, even though the BAU scenario assumes no specific policies. Given this, a time frame that encompasses these events is preferred, so that the BAU is not limited to a 'no-growth' or 'no-prosperity' scenario.

2.3.2. Net-zero scenario

On the other hand, the net-zero economy scenario assumes policies and actions are taken to change the economy's structure. It is assumed that the policies and actions proposed in the NDP and the TES are implemented efficiently. These implementations would produce both changes on the final demand and changes on the interactions between economic activities, *i.e.*, changes on the intermediate demand. The relations between economic activities that are not described in the NDP or the TES are assumed to remain constant, and that their growth follows its historical trend.

In order to identify the changes introduced by the transformation of the economy, several sources of information were used. The main source is the Climate, Land, Energy and Water regional model (CLEW-Regional-CR; Quirós-Tortós et al., 2021) which studies the interaction between the supply and demand of different economic activities in the regions of Costa Rica. This model has included the NDP policies to analyze the costs and benefits of the implementation of the country's long-term strategy, which provides innovative and highly relevant information to this study.

In general, CLEW models are tools for simultaneous consideration of food, energy and water security and allow to assess how production may contribute to climate change. Additionally, through these models it can be identified how climate change may affect resource systems. For this specific reason, CLEW models are recommended for estimations regarding climate change and climate action (Beltramo et al., 2021; Sridharan et al., 2020), and have been used in several case studies (Howells et al., 2013; Pereira Ramos et al., 2020, 2022). For Costa Rica in particular, the model has been used to estimate the pathways to decarbonization (Roccard Pommera and Victor-Gallardo, 2020; World Bank, 2020). Therefore, CLEW-Regional-CR (Quirós-Tortós et al., 2021), which considers the NDP actions, results the best estimate available regarding changes in economic activities due to the policies to be implemented. This model, however, considers precipitation and temperature changes to estimate hydrological needs, but does not include IPCC projections of climate events such as El Niño.

Alongside this source, macroeconomic indicators from the Central Bank of Costa Rica, demographic information from the National Institute of Statistics and Censuses, investment data from the Costa Rican Institute of Railways, and detailed policies and actions from other Costa Rican national long-term plans (energy, transport, waste management, composting, cattle raising, etc.) were used. These data, specific to Costa Rica, allows to project what the investment requirements will be, and what pressures from demand would fall upon which economic activities. Given this information, the changes in the dynamics between sectors can be estimated, so that a net-zero scenario can be derived.

Taking this information, each component is weighted by their participation on the final demand to estimate the changes on the final demand (Schuschny, 2005; Venegas, 1994). By doing so, the components' variation is isolated from the rest of

the final demand, and their impacts over the required gross production are estimated separately.

Furthermore, changes on the intermediate demand follow the approach displayed where the variations on the relations of the components are applied as percentage impacts on the initial interaction matrix (Bacharach, 1970; Parikh, 1979). If the initial coefficient matrix A_0 , that proxy the technology of the economy, is taken, and assuming intertemporal changes as the substitution and production effects, a new matrix $A_1 = r \times A_0 \times s$ is obtained, where s and r are diagonal matrices that represent the production and substitution effects, and that are determined by a proportionality factor.

Since these changes on the intermediate demand have an effect on the coefficient matrix, employment and output multipliers are recalculated every period. For the case of this study, due to the fact that policies and actions are expected to be progressively taken every five years, as stated in the NDP and TES, different multipliers are used per quinquennium.

2.4. Assessing uncertainty of policies

The estimation is initially done considering baseline assumptions over the evolution of intermediate and final demands in a transition to a net-zero economy. These assumptions are based on the best available and validated information. However, uncertainty is large, especially when studying the long-term evolution of the economy in a country.

To examine the effect of uncertainty on the estimations, an analysis based on a robust decision-making (RDM) approach (Lempert, 2019) is taken. One hundred versions of the scenarios were simulated, each one varying the input parameters in a unique way. There are five categories of input parameters: demands of the economic activities, technical coefficients, jobs per unit of output, household unitary consumption and unitary wages. Each one of the one hundred simulations is called a future (as per usual RDM terminology). Each future is designed such that the scenarios are comparable, differing only by the modeled effect of net-zero transition policies.

This study takes a Latin hypercube sampling (LHS) approach to vary the input parameters. The LHS is used to produce one hundred unique combinations of values between 0.9 and 1.1 for each parameter; and the one hundred combinations are equivalent to the one hundred futures. The values are scaling factors that multiply each parameter; the original values of the parameters are the ones used in the scenarios under baseline assumptions. The values were taken between 0.9 and 1.1 so that the deviations of the parameters with respect to the baseline assumptions resulted in $\pm 10\%$ or less, which is the prevailing range of variation of the parameters in the Costa Rican economy for the last twenty years.

Additionally, with the LHS approach the demand for the net-zero economy scenario varies relative to the demand for the BAU scenario by using a ratio of the demands between scenarios. In other words, for each net-zero scenario with variations there is also a BAU scenario with variations in the same direction, so that these are comparable. It is important to

note that the ratio used in the demands is interpolated to smooth the resulting change (with respect to the first five years); in the first five years the demand is multiplied by one, and in the last five years by the ratio.

3. Results

3.1. National level

Under baseline assumptions, a net-zero economy would create 135 thousand net jobs by 2050 in the entire country (Fig. 1), which represents 2.7% of the Costa Rican population and 7% of the employed population in 2017.

From these additional 135 thousand jobs in the net-zero economy, around 23 thousand would be directly created from transformative actions taken to reach the net-zero economy, while more than 70 thousand would be created indirectly, due to the economic linkages and chains of production of the country. About 42 thousand of the additional jobs by 2050 would be induced, *i.e.*, would be created due to the dynamization that the new direct and indirect jobs would bring to the economy.

The results show that the additional jobs will be obtained in an increasing way as the policies and actions are being implemented. In the short term (2021–2025), when the policies are primarily preparatory and organizational, around 11 thousand net jobs would be created. In the medium term (2026–2035), this number would increase five times and, specifically during the 2031–2035 period, the greatest increase of jobs (43 thousand) would result from the deployment of policies and actions for the transformation of the economy embodied in the NDP and the TES. In the long term (2036–2050), the actions of these plans would materialize, and more than 65 thousand net jobs would be created.

The results suggest that sustainable activities, *i.e.*, activities related to electrification, renewable energy, digitalization, etc., will create more jobs. Activities related to clean energies grow the most and provide most of the net jobs (Fig. 2). The utilities sector along with the commerce, transportation, and information and communication sectors account for 84% of the total net jobs.

The utilities sector, that includes all forms of electricity generation, water provision and treatment, and sanitation

services, would be the one to gain the most additional jobs: a total of 54 thousand by 2050. Many of the actions proposed by the NDP revolve around the electrification of public, private and cargo transport, as well as industrial production, and the plan seeks full national coverage of sanitary sewerage and wastewater treatment, which would explain the creation of 35 thousand direct and indirect jobs by 2050. Additionally, the induced jobs would result due to the digitalization and electric energy and sanitation needs in all the other sectors of the economy and households.

Activities related to commerce, including wholesale, retail and repair of vehicles, would increase in almost 30 thousand net jobs by 2050. The activities related to wholesale and retail commerce would increase the number of net jobs in almost 27 thousand, as a result of their high economic linkage with many of the activities impacted by the transition to a net-zero economy, as well as the changes in the living standards of the population and increased mobility. Although the electromobility promoted by the NDP would reduce the costs of maintenance and operation of electric units (Quirós-Tortós et al., 2019), the country would require a skilled workforce for the maintenance and repair of the electric private, public and cargo fleet, so that around 3 thousand net jobs would be gained, since that is a niche that currently does not exist in the country.

Activities related to the transportation sector would create 16 thousand additional jobs, 13 thousand of which would be direct and indirect. The increase in transportation activities is expected due to higher use of public transport, which would be a consequence of the deployment of more means of transport and its increased efficiency. Another reason for higher transport activity is the development of electric freight transport and investment in transport infrastructure. The information and communication sector would generate almost 14 thousand net jobs by 2050 under a net-zero economy scenario, as a result of the transformation towards a digitized economy, which implies expanding the population's access to telecommunications and the development of these economic activities.

However, the transition to a net-zero economy also means an increase in regulations, as well as technology improvements that might affect labor-intensive activities. In Costa Rica, this is the case for the activities related to agriculture and

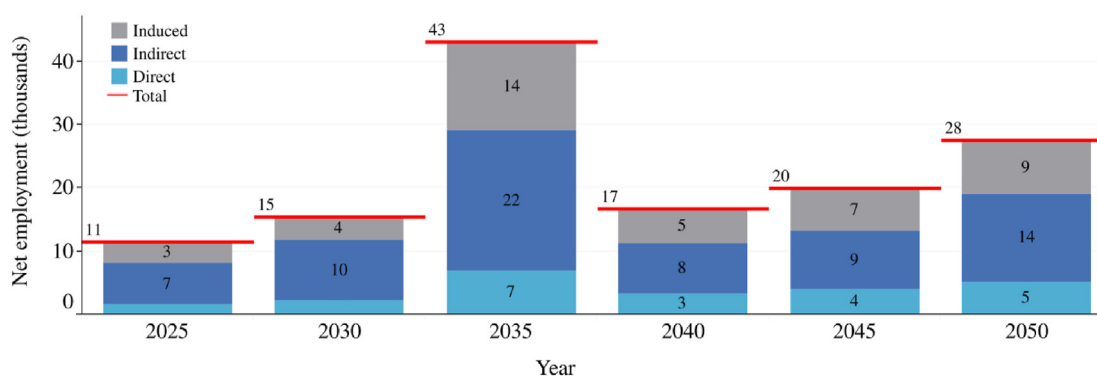


Fig. 1. Direct, indirect and induced net jobs per period at a national level, 2021–2050.

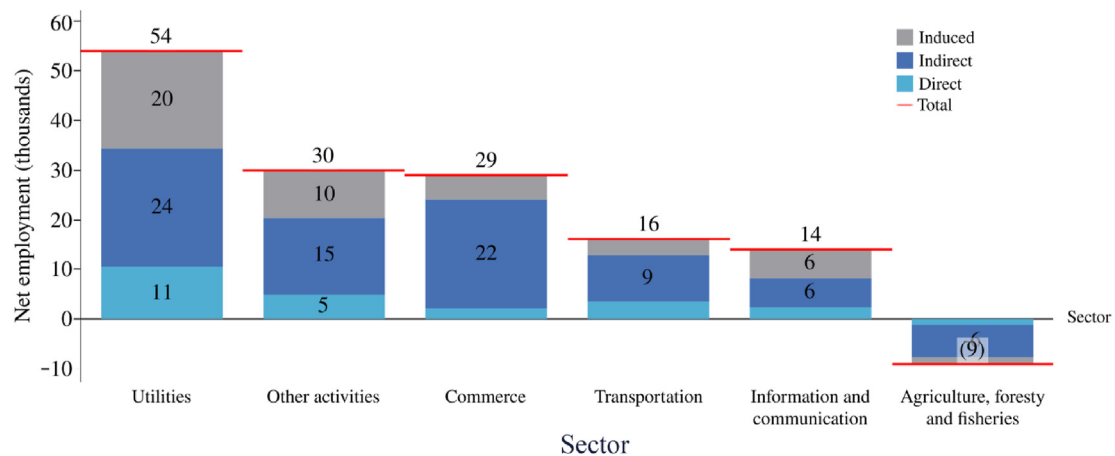


Fig. 2. Direct, indirect, and induced net jobs per economic activity at a national level, 2021–2050.

mining, which the results show that would create fewer jobs by 2050 under the decarbonized scenario. Particularly in agriculture, the deployment of new technologies, the reduction in land coverage requiring less workers, the improvement in productivity, and the transition to better and sustainable practices in the livestock sector would require more qualified but lower number of jobs than those needed under a BAU scenario. It is estimated that under the BAU scenario, by 2050 the agricultural activity would require 95 thousand new jobs, while under a net-zero economy scenario, the number would be 86 thousand. Additionally, it is expected that, although in the intermediate period there would be requirements of highly qualified labor in the agriculture sector, in the long run there would be a drop in the overall labor requirement, as new technologies and substitution practices are put in place. In the case of mining, the regulations over this activity are expected to be broad and strict as a result of decarbonization in this process. The reduction in jobs, however, is estimated to be less than 300 by 2050, since the mining sector is relatively small, as it only employs 0.2% of the working population. Mining activities in the country are related to extraction of stone, sand and clay. Regarding fossil fuels, the country does not extract them, so that it is a gross importer.

Regarding the composition of jobs, the model and data, alongside information from Costa Rica's National Household Surveys, allow to estimate the impacts of the transition in terms of jobs over specific populations. The main results of these estimations show that the transition to a net-zero economy would not only benefit the working population in general, but also increase formal employment in vulnerable populations. In Costa Rica, women, people with disabilities, migrants and indigenous people are underrepresented in the labor market, and historically have had the highest rates of underemployment, unemployment and poverty. The results suggest that the implementation of the NDP and TES would impact female labor in a greater magnitude than male labor; net jobs would be 1.3 times relatively higher for women than men, considering the current disparities. Additionally, the underrepresentation gap in employment between indigenous

population and non-indigenous population in the country would be reduced by 5 percentage points; for migrant population, this gap would be reduced by 8 percentage points.

Nonetheless, based on the current composition of the economic sectors and their historical trends, together with the current orientation of human capital formation by sector, not all the jobs that the transformation encourages will likely be filled. Policies regarding gaps in the formation of skills should be taken, so that the benefits of the transition can be reaped.

3.2. Regional level

The results are analyzed at the Costa Rican regional (sub-national) level. Costa Rica is divided into six socioeconomic planning regions, which are diverse in climate, energy availability and needs, land, and production. Under baseline assumptions and with respect to the BAU scenario, all the regions would see an increase in jobs in a net-zero economy. Net jobs per region vary between 8 and 35 net jobs per thousand inhabitants in 2017, depending on the production structure in each of them (Fig. 3).

Costa Rica's Central region is located in and around the Central Valley, and it is the economic and political center of the country. It concentrates more than 60% of the country's population and around 70% of its GDP. This area has faced the greatest industrial development in the territory. The results show that under a net-zero scenario at least 35 net jobs would be generated per thousand inhabitants in 2017, making it the region that would gain more from the transition. The activities with the greatest impact on job creation in this region would be the ones related to the utilities, the commerce, the transportation and the information and communications sectors, as well as professional services, health care, education, and finance. Given the cluster of human capital in the region, a shift in the offer of skills training (formal and informal education, *etc.*) is especially necessary to secure a just transition and a fulfillment of these jobs.

The Chorotega region is located at the northwest of the country and is the main producer of electric energy in the

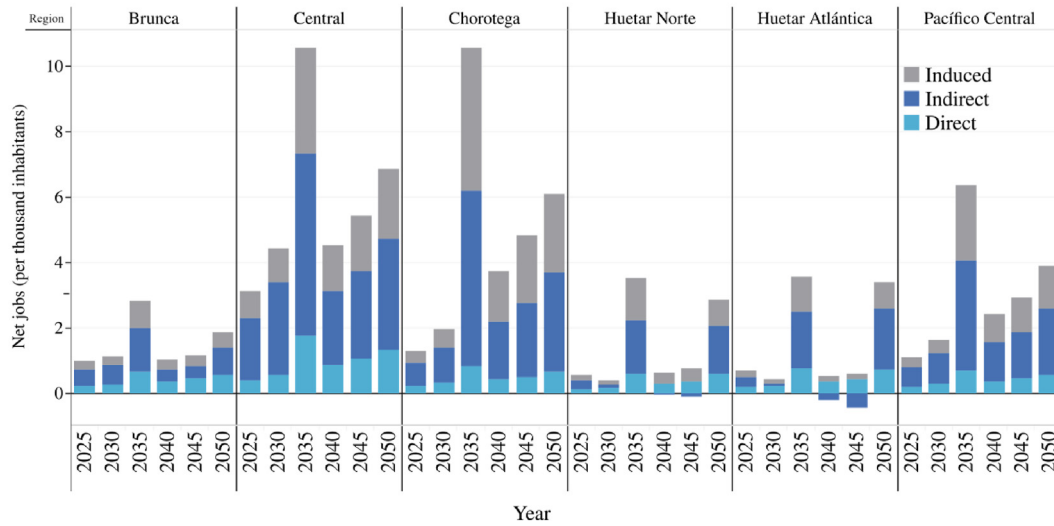


Fig. 3. Direct, indirect, and induced net jobs per period and thousand inhabitants at a regional level.

country; it is filled with wind farms, due to its location in the wind corridor of Central America, and geothermal power stations, due to its proximity to volcanoes. This region is the second most benefited from the transformation by 2050: it would gain 29 more jobs for every thousand inhabitants in 2017. The majority of the net jobs would come from the utilities sector, but also from commerce and manufacturing. In other words, the region would gain jobs if it exploits its vicinity to natural sources of clean and renewable energy.

The Pacífico Central region is the smallest in the country; it is located at the coastal plain of the Pacific Coast. This region has beaches and transitional tropical forests that make tourism and coastal activities the main economic activities. Pacífico Central would benefit from about 18 additional jobs for every thousand inhabitants in 2017 in a net-zero scenario. Most of the additional jobs would be obtained from the utilities sector related activities. Nonetheless, this region would be the most affected by the regularization of mining and quarrying, facing a negative effect jobwise of less than 200. Policies to move employed people from the mining sector to the utilities sector in the region should be considered.

The results suggest that in the Brunca, Huetar Atlántico and Huetar Norte regions between 8 and 9 additional jobs would be created for every thousand inhabitants in 2017 of the corresponding regions in a net-zero scenario. The Brunca region is located in the south of the country, and it is the rainiest area. It also has a long mountain range and a coast. The region is home to most of the country's indigenous population and depends strongly on agriculture. The activities that would obtain more jobs from a net-zero scenario would be those related to commerce. The Huetar Atlántico region is located at the Atlantic Coast and it possesses the main port in the country. In this region, the activities that would gain most jobs would be from the utilities and commerce sectors. Since the region depends on agriculture, specifically bananas, boosting their production would increase the benefit in terms of net jobs.

Regarding the Huetar Norte region, it is located in the north of the country, and it is a very flat area with very fertile land, so there is a large presence of extensive agriculture and cattle raising. This region would follow the national trend on net jobs obtained, although in relative terms the manufacturing industry would be one that would grow the most per thousand inhabitants. Similar to what is concluded in Wang et al. (2023c), local governments should make full use of their administrative power to stimulate the development of environmentally sustainable industries and activities, so that a 'win-win' outcome is landed. These positive effects on net jobs can be obtained if the stimulus and promotion of actions is in place.

When analyzing the cantons specifically, the results show that most of them would increase the number of jobs: 78 of the 81 cantons would have a higher job growth rate under a net-zero economy scenario than a BAU one. The cantons that would obtain the most jobs per thousand inhabitants are those related to the generation of clean electricity through wind farms, solar farms, geothermal plants and dams. Cantons that are strongly related to commerce, transportation and information and communication activities will also have an increase in the number of jobs by 2050. The three cantons that would not benefit from a net-zero economy are mainly dependent on agricultural activities, currently labor intensive, which makes the optimization of the production method reduce the job growth rate. For these specific cantons, considering this impact on time might allow to restructure some of the productive and labor matrix before 2050, so that negative effects might be mitigated.

3.3. Results under uncertainty

The previous results use baseline assumptions based on the best available information for both the net-zero and BAU scenarios. However, it should be considered that this

information is subject to uncertainty, so the effects of variations in demand, technical and wage coefficients, household consumption and employment per output should be incorporated. Uncertainty over these variables was explored using an analysis inspired by the robust decision-making method (RDM). A total of 100 plausible implementation paths for each scenario were designed for a total of 101 unique parameter combinations per scenario.

The results of the analysis show that a net-zero economy scenario would generate more jobs than the BAU scenario in all cases (Fig. 4). Favorable conditions, such as effective and timely policy implementations, as well as an eased rearrangement of the workforce, could cause the number of net jobs to reach 750 thousand, more than five times the net jobs obtained in the scenarios with baseline assumptions (*i.e.*, baseline net-zero and BAU scenarios); under these

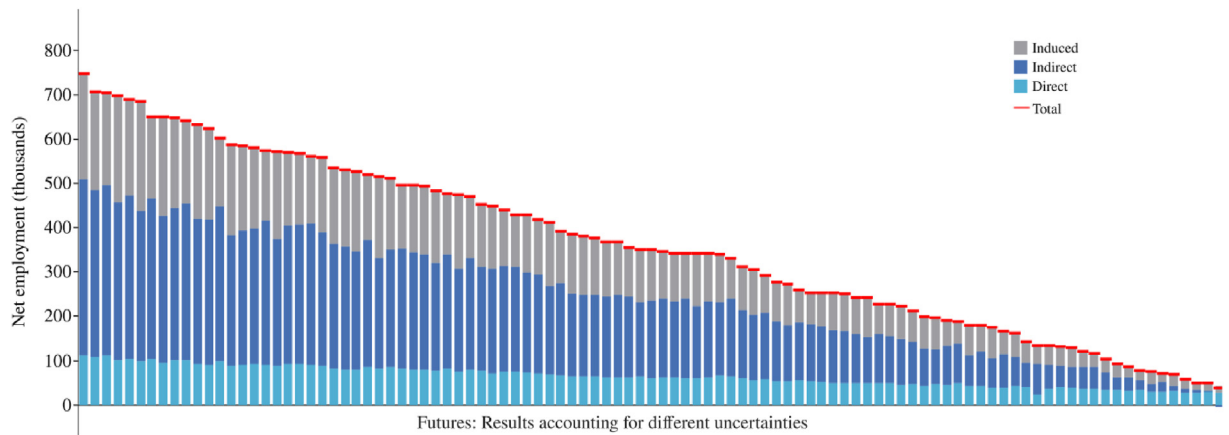


Fig. 4. Direct, indirect, and induced net jobs per future at a national level.

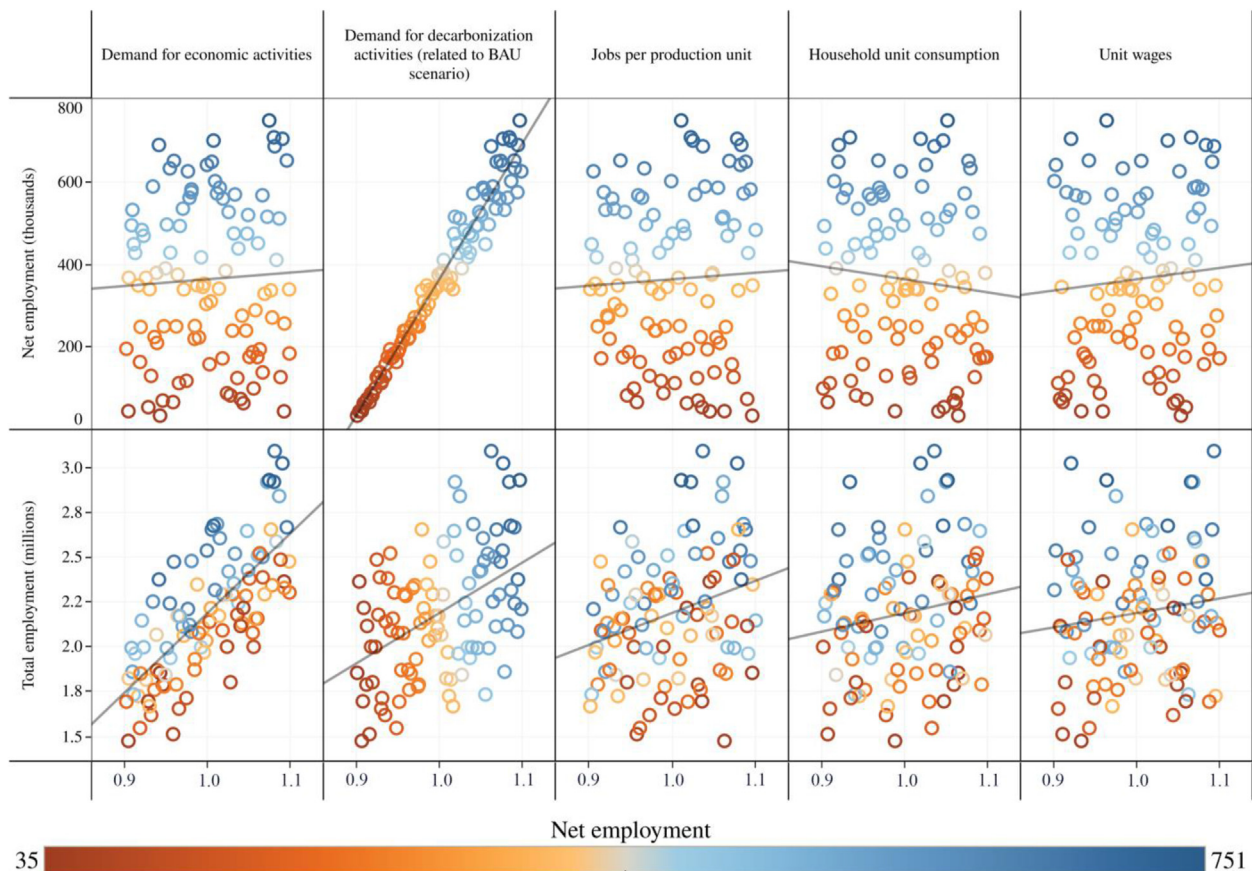


Fig. 5. Effect of input parameter variation on the number of net jobs and total jobs in net-zero scenarios.

circumstances, the unemployment rate in the country would be reduced significantly and the participation rate would increase. In addition, more than 80% of the net-zero futures lead to more net jobs than the net-zero under baseline assumptions; in other words, over 135 thousand net jobs by 2050. The least favorable implementation path, where policies are delayed or untimely, or the workforce faces skill mismatches, would result in about 35 thousand net jobs by 2050, which represents a 1.8% of the employed population in 2017 in Costa Rica.

As in the baseline assumptions net-zero economy future, it is concluded that the 2031–2035 period is the one with the greatest job creation in the majority of the paths. That is, even when incorporating future uncertainty, policy actions in this period would provide the economy with greater opportunities for economic growth, and therefore greater job creation. Additionally, the parameters that determine the number of jobs obtained were found to be the demands of economic activities, technical coefficients (relations between economic activities), jobs per unit of production, unit household consumption, and unit wages. The economic sectors where the largest differences between the technical coefficients and demands are mainly observed are agriculture, construction, and transportation, which are also the most labor-intensive.

The demand for each economic activity associated with the net-zero economy may vary relative to the demand for the same activity in the scenario without the net-zero economy. The relative demand for these economic activities is the parameter that causes the largest variation of net jobs (Fig. 5). This can be observed with the Pearson correlation coefficient between changes in relative demand and net jobs, which is 0.98. The other parameters do not have a significant impact on net jobs. However, when considering total employment, it is the demand for all economic activities that has the greatest impact, with a coefficient of 0.73 ($p < 0.05$). The other parameters have a greater impact on total jobs than on net jobs.

These results imply that the driver for job creation under these circumstances is the demand for the economic activities related to sustainability and decarbonization. Investment in these activities and a change in consumption patterns would increase jobs overall. As an example, if the unit of demand for sanitation services is doubled, then jobs would increase by 2.5 and this change would be driven by the new economic linkages (both backwards and upwards) of the economy.

4. Conclusions and discussion

4.1. Conclusions

Under baseline assumptions, a net-zero economy would produce 135 thousand more jobs by 2050, which represents around 7% of the employed people in Costa Rica in 2017 and 6% of the labor force. The baseline assumptions result seems to coincide with the estimates for Latin America and the Caribbean (PNUMA, 2020; Saget et al., 2020), which suggest that the transition towards clean energy and electric mobility would generate a job increase that represents around 10% of the regional labor force.

Additionally, the baseline assumptions result shows that only one fifth of these 135 thousand net jobs would be directly generated, while the others would result from the economic linkages and dynamization of the economy. When considering that many world-wide and nationwide estimations have been focused on the impact of the transition on direct jobs only (IRENA and ILO, 2021; Ram et al., 2022; Ravillard et al., 2021), this result is relevant as it might incentivize more full-scale and prompt transitions since the positive impact in jobs can be bigger than initially estimated. Moreover, the estimations contemplate national decarbonization and development strategies, that impact all of the economy in different sectors and in several ways. Most of previous studies have focused on particular policy or impact area, where this study focused on all of the economy and all of the policies.

Furthermore, it is found that economic activities related to clean energies would have higher employment growth rates, while activities that become more regulated would have lower rates. This is congruent with literature that estimates the employment factor between renewable investment and investment in fossil fuel (Garrett-Peltier, 2017). It is also found that this result holds when including uncertainty, as relative demand for these activities has a large impact on job variation. However, it should be noted that these results do not consider the role of national imports and exports, so that it might be possible that in some activities there will be some substitution effect.

The results show that regions that are most linked to renewable energy generation and digital services related activities are the ones that gain the most jobs; the gains of the transition to a net-zero economy are distributed unevenly across the country. Although the specific magnitude of the effect of the transition is difficult to estimate due to the model not being able to quantify trade and the ease of mobility between regions, the direction of the effect is clear.

Further, the results present that labor-intensive activities have the widest changes between their demand and their relations to other activities. In order to generate effective policy and achieve a just transition, these disparities should be considered (Sonja and Harald, 2018).

It is shown that even when including uncertainty over policy magnitudes, net-zero scenarios create more jobs than the BAU ones, and that the most important driver of net jobs is the relative demand of economic activities associated with a net-zero economy. Due to the lack of studies around intensities and uncertainty (Ackerman et al., 2009; Beck and Krueger, 2016; Pindyck, 2017), these seem especially relevant, as they present the different pathways to decarbonization that could be taken. Finally, including uncertainty and different possible futures makes it more trustworthy for policy-makers and stakeholders to decide to take action, and this can by itself accelerate and increased the gains from the transition.

4.2. Policy implications

In general, the implementation of Costa Rica's NDP and TES would impact positively the economy in terms of jobs.

The main result by itself should promote the application of these policies, and support the continuation of climate action.

Nonetheless, these gains will not be obtained automatically. Further studies on the technical and professional requirements that the new jobs would have, as well as needed investments in education and infrastructure are necessary. Specifically, around the sectors with the most changes, such as energy, transportation and agriculture, policies regarding capabilities should be taken. Future labor demand needs should be considered from day one of the implementation, as skill formation from the labor supply takes time. For example, education policies should be taken towards ‘jobs of the future’, increasing funding and supply for these. Additionally, the public sector should generate agreements with the private sector, particularly for the more technical jobs, so that education and training programs can be developed.

Furthermore, actions towards minimizing the negative net jobs in economic activities should be also considered, so that a just transition is guaranteed. In regions particularly impacted by the degrowth of the agricultural sector, policies regarding investment in technological advances and their adoption, as well as restructuring of the productive matrixes might help mitigate these effects. Moreover, regions that will not be subject to high levels of employment growth should invest in infrastructure and projects that could allow them to reap more gains related to the transition.

4.3. Limitations and future recommendations

Although this study sheds light on the employment implications of the transition to a net-zero economy in Costa Rica, there are still some limitations. First of all, the input–output model uses data from a particular period, and information of changes over time is not available; for the Costa Rican case, the latest I–O matrix is from 2017. Therefore, despite it being the best available information for the analysis, it might not represent perfectly the structure of the economy, particularly considering the impact that the Covid-19 pandemic had on it. Another limitation is that the calculation of technical coefficients assumes these are fixed, so that it implies the relationship between sectors does not change over time and that productive sectors have constant returns to scale. If a more recent I–O matrix is made available, future studies should incorporate the dynamics of the change in the estimations, considering these periodic variations as part of the business-as-usual trends.

As mentioned previously, the model does not consider the role of trade, so that international substitution effects cannot be quantified. Following studies on the topic should analyze the possible fluctuations in commerce and their effect on the economy. Additionally, the transition to a net-zero economy in Costa Rica is subject to an increase in electromobility, which can have negative environmental implications, if considered the impacts of mining for lithium, for example. Even though this study focuses on Costa Rica, and the country is not a producer of batteries, these effects should be considered when presenting the transition as sustainable.

Finally, this analysis considers the NDP and TES as is, however, Long-Term Strategies are subject to constant change and actualizations, so the effects can be widened or shortened depending on these adjustments. Furthermore, the impacts of climate change on the Costa Rican economy were not included in the modelling, so that investments or policy requirements might be needed in the future, which would then vary the estimates. Although the final number of net jobs cannot be known ex-ante, considering the changes and their potential effect on the labor market is relevant, so that timely policies regarding human capital are taken.

Declaration of competing interest

The authors declare no conflict of interest.

CRedit authorship contribution statement

María José Sauma Chacón, Alejandro Matarrita Valverde and Bernardo Zúñiga Alvarado worked on the methodology, software, data curation, validation, formal analysis, writing and visualization of the manuscript. Mónica Rodríguez-Zúñiga and Jairo Quirós-Tortós provided the conceptualization, resources, administration and supervision of the project, and reviewed and edited the text. Luis Víctor-Gallardo, Pauline Schaal, Paula Cubillos, Thierry Duplan, Matthieu Robin and Slim Dali reviewed and edited the text. All authors gave their final approval to the manuscript.

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